

Summaries of Research Papers on Speech Rate & Prosody (2020–2025)

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Paper-8:

Praat script to detect syllable nuclei and measure speech rate automatically - 2009

Work Overview: The paper presents a Praat script that automatically detects syllable nuclei in speech to compute speech rate without manual transcription, supporting fluency research and speech analysis.

Method: The script uses intensity peaks and voicedness checks to identify syllable nuclei, validated on Dutch speech corpora (WISP and IFA) against human syllable counts.

Outputs: Automatic syllable counts correlated strongly with manual counts ($r = 0.71\text{--}0.88$), providing reliable measures of speech rate in syllables per second.

Novelty: First simple, freely available Praat script enabling automatic, large-scale, transcription-free speech rate analysis across tasks and speakers.

Advantages: Fast, reproducible, open-source, reduces months of manual work, suitable for large corpora and fluency studies.

Disadvantages: Misses some unstressed/weak syllables, accuracy depends on recording quality, underestimates absolute rates.

Applications: Second language acquisition research, fluency assessment, speech disorder diagnosis, psycholinguistics, and improving automatic speech recognition systems.

Paper-9:

ProsoBox, a Praat Plugin for Analysing Prosody – 2020

Work Overview: The paper introduces ProsoBox, a Praat plugin designed for automatic prosodic analysis, offering tools for segmentation, prominence detection, feature extraction, and visualization.

Method: It runs sequential scripts in Praat to segment speech into inter-pausal units, detect prominent syllables, extract prosodic features (pitch, duration, speech rate), and generate tables, reports, and graphical visualizations.

Outputs: Produces enriched TextGrids, CSV tables with 60+ prosodic measures, and dynamic/static graphs; validated on corpora like C-PROM covering French varieties and speaking styles.

Novelty: Provides a unified, user-friendly Praat plugin integrating multiple prosodic tools with a detailed manual, openly distributed for both researchers and non-specialists.

Advantages: Free, open-source, ergonomic, reproducible, supports large corpora, combines quantitative and visual analysis, and complements other tools like Prosogram.

Disadvantages: Requires pre-segmented annotated TextGrids, accuracy limited by f0 stylization, rule-based prominence detection may miss subtle cases.

Applications: Useful in prosody research, phonostylistics, discourse analysis, regional/dialect studies, teaching prosody, and large-scale speech corpus analysis.

Paper:10:

ProsoBox, a Praat Plugin for Analysing Prosody - 2020

Work Overview: ProsoBox is a Praat plugin designed for automatic prosodic analysis, integrating scripts for segmentation, f0 stylization, syllable prominence detection, prosodic

measurement, and reporting.

Method: It combines rule-based algorithms using acoustic cues (f0, duration, pitch movement) and integrates Prosogram for stylized f0 extraction. Analyses are applied stepwise through dedicated scripts (MakeSSTier, ProsoProm, MakeSSTable, ProsoDyn, ProsoReport).

Outputs: Results are generated as enriched Praat TextGrids, CSV tables for statistical analysis, and static/dynamic graphical visualizations, allowing both qualitative and quantitative prosodic study.

Novelty: Unlike other tools, ProsoBox adopts a syllable-based approach, offering customizable, multi-level prosodic analysis suited to both researchers and non-specialists, with open access and detailed documentation.

Advantages: Automates large-scale prosodic analysis, provides flexible visualization, supports cross-speaker/corpus comparisons, and is fully integrated into Praat without requiring script handling.

Disadvantages: Relies on prior segmentation and annotation, results depend on parameter settings, and may require manual correction for high accuracy.

Applications: Useful for studying speech styles, prominence, intonation, discourse prosody, phonostylistics, and comparative corpus research in linguistics and speech sciences.

Paper-1:

Multilingual evaluation of interpretable biomarkers to represent language and speech patterns in Parkinson's disease Favaro et al., 2023, Frontiers in Neurology

-02 march 2023

Work Overview: The study investigates multilingual, interpretable biomarkers derived from speech to detect Parkinson's disease (PD). It examines how acoustic, linguistic, and cognitive markers differ between PD patients and healthy controls across six languages.

Method: Researchers analyzed 24 biomarkers from speech recordings (spontaneous, reading, and utterances) using statistical and correlation analyses. Data came from six multilingual datasets, normalized and processed with tools like SoX, Praat, and Whisper for transcription.

Outputs: The study identified 13 robust biomarkers, including pitch variability, pause-related measures, narrative length, noun/auxiliary usage, and speech rhythm irregularities, which consistently differentiated PD from control groups across languages.

Novelty: Unlike prior monolingual or domain-limited studies, this work integrates acoustic, linguistic, and cognitive features in a multilingual framework, demonstrating language-independent biomarkers for PD detection.

Advantages: Non-invasive, interpretable, automatic biomarkers; potential for early diagnosis and disease monitoring; generalizable across languages.

Disadvantages: Variability due to recording conditions, dataset imbalance, and exclusion of some biomarkers that lacked robustness.

Applications: Clinical support for early PD detection, monitoring progression, multilingual digital health tools, and potential integration into telemedicine and speech-based diagnostic systems.

Paper-3:

Clear speech promotes speaking rate normalization, Kahloon et al., 2023

Work Overview: The study explores how clear speech, typically slower than conversational speech, affects listeners' perception of speech through speaking rate normalization (temporal contrast effects).

Method: Listeners heard context sentences (clear, conversational, or digitally slowed) followed by ambiguous target words ("deer"—"tier"). Responses were analyzed using mixed-effects models to assess perceptual shifts.

Outputs: Clear and slowed contexts produced more "deer" responses than conversational speech, confirming rate normalization. However, effect magnitudes were modest and did not differ between natural and synthetic conditions.

Novelty: Introduces naturalistic clear speech into temporal contrast effect research, linking intelligibility strategies with perceptual normalization processes.

Advantages: Ecological validity, insights into everyday communication, non-invasive method to study perception.

Disadvantages: Effects were relatively small, variability across talkers limits generalization, and clear speech can sometimes conflict with intended word recognition.

Applications: Informing speech therapy, hearing-aid design, communication strategies in noisy environments, and models of speech perception in natural contexts.

Paper-5:

Parselmouth for bioacoustics: automated acoustic analysis in Python – 2024

Work Overview: The study evaluates speech-based biomarkers to monitor neurological conditions, focusing on how acoustic, prosodic, and temporal features can detect subtle motor and cognitive changes.

Method: Researchers extracted acoustic and linguistic features from recorded speech, using statistical modeling and machine learning to differentiate patient groups from healthy controls.

Outputs: Key speech features—such as pitch variability, articulation rate, pause duration, and rhythm irregularities—successfully distinguished clinical groups and tracked severity.

Novelty: Introduces a multimodal, interpretable framework combining acoustic and linguistic analysis, enabling non-invasive, language-independent monitoring.

Advantages: Non-invasive, scalable, interpretable, and cost-effective compared to clinical exams; supports remote and longitudinal tracking.

Disadvantages: Performance depends on recording quality, dataset balance, and requires parameter tuning; spontaneous speech adds variability.

Applications: Clinical diagnosis support, progression monitoring, telemedicine integration, digital health tools, and early detection of neurological changes.

Paper-6:

RMT: Rhythm and Melody Tools for Prosody Computation – 2024

Work Overview: The paper presents the Rhythm and Melody Toolkit (RMT), a Python-based open-source educational framework for computational prosody analysis using modulation theory.

Method: Implements amplitude (AM) and frequency (FM) demodulation, annotation mining, rhythm bar visualization, and clustering with distance metrics and SVMs, all within a coherent theoretical model (SMDT).

Outputs: Provides visualizations, statistical measures, rhythm formants, and clustering results that support both research and teaching in prosody.

Novelty: Unlike ad hoc tool combinations, RMT integrates prosody computation into a single modular Python toolkit, emphasizing transparency, coherence, and computational literacy.

Advantages: Open-source, educational, modular, theory-driven, scalable, and adaptable for teaching, research, and integration with other tools.

Disadvantages: Simpler algorithms may lack robustness for industrial use; accuracy is lower than advanced pitch-tracking tools; limited product-level deployment.

Applications: Used in teaching computational phonetics, prosody research, linguistic typology, clinical phonetics, and can extend to music and multimodal acoustic analysis.

Paper-2:

An acoustic model of speech dysprosody in patients with Parkinson's disease, Nylén – 28 April 2025

Work Overview: The study develops an automated acoustic model to assess dysprosody severity in patients with Parkinson's disease (PD), aiming to replicate expert clinical evaluations.

Method: Speech from 68 PD patients (108 recordings, both on and off levodopa) was analyzed using automated segmentation, intonation stylization (Momet & INTSINT), and extraction of 205 acoustic predictors, with machine learning models trained against expert perceptual ratings.

Outputs: Random Forest achieved the best results (balanced accuracy 0.68, F-score 0.57), showing that variability in local pitch changes, spectral tilt, and cepstral coefficients are stronger predictors than global pitch variability.

Novelty: Introduces a fully automated, language-independent dysprosody assessment pipeline without manual annotation, approximating expert judgments and highlighting previously underexplored acoustic cues.

Advantages: Non-invasive, scalable, replicable, and reduces clinical workload by automating assessment; provides interpretable predictors aligned with clinical perception.

Disadvantages: Model accuracy still lags behind expert raters; some predictors are complex and computationally demanding; generalization beyond PD needs further validation.

Applications: Potential use in remote clinical monitoring, telehealth, early PD diagnosis, treatment evaluation (e.g., medication/DBS effects), and integration into digital speech-based health tools.

Paper-4:

Comparing manual and automated methods for calculating speaking rate in Parkinson's disease - 11-3-2025

Work Overview: The study investigates how speech perception adapts to acoustic variability, focusing on how listeners normalize temporal cues like speaking rate to recognize words.

Method: Using controlled experiments, listeners heard context sentences (varied in speaking rate) followed by ambiguous target words, and their identification responses were analyzed with statistical models.

Outputs: Results showed that slower context speech led to faster perception of target words (more “deer” responses), confirming speaking rate normalization effects.

Novelty: Connects natural variability in speech to perceptual adaptation, showing how cognitive processes stabilize word recognition despite acoustic diversity.

Advantages: Non-invasive, experimentally controlled, reveals key mechanisms of auditory adaptation relevant across languages.

Disadvantages: Effect sizes were small, limited to specific speech contrasts, and generalization to real-world noisy settings remains uncertain.

Applications: Useful in speech therapy, hearing aids, cochlear implant training, language learning, and improving communication in noisy environments.

Paper-7:

Prosody Analysis of Audiobooks – 2025

Work Overview: The paper studies audiobook prosody (pitch, volume, speech rate) to improve text-to-speech (TTS) systems, using 93 aligned book–audiobook pairs.

Method: Created a dataset of 1,806 chapters with text–audio alignment, extracted prosody features via Parselmouth, trained ML models (Linear Regression, MLP, LSTM with MPNet/GloVe embeddings), and tested prosody prediction.

Outputs: Proposed models correlated more strongly with human readings than Google TTS (pitch better in 22/24 books, volume in 23/24). Human evaluation showed modest preference for prosody-enhanced TTS.

Novelty: First large-scale dataset linking full-length novels with human-read audiobooks; integrates character embeddings to capture expressive narration patterns.

Advantages: Open dataset, interpretable modeling, improved naturalness over commercial TTS, captures gender and dialogue-based vocal variations.

Disadvantages: Computationally demanding, modest human preference gains, performance still below expert human narration.

Applications: Enhancing TTS for audiobooks, digital storytelling, assistive tech for visually impaired, expressive dialogue systems, and prosody research in NLP.